
DIGITAL ECONOMY IN NEPAL: OVERVIEW OF POLICIES, ETHICS, AND FUTURE SKILLS

DIGITAL ECONOMY

- Economic activities using digital tech
 - Includes payments, e-commerce, AI, cloud, 4IR tech
 - Global example: Amazon, Alibaba
 - Nepali example: eSewa, Khalti
 - Impact: Faster, cost-efficient transactions
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IMPORTANCE FOR NEPAL

- Digital adoption drives economic growth
 - Reduces dependency on cash transactions
 - Supports startups & innovation
 - Enables remote work & outsourcing
 - Example: Online tutoring in Nepal during COVID
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POLICY & REGULATORY IMPLICATIONS

NEPAL RASTRA BANK (NRB) POLICIES

- Guidelines for mobile wallets (Khalti, eSewa)
 - KYC mandatory for transactions
 - Transaction limits to reduce fraud
 - QR interoperability across wallets
 - Example: Pay via Khalti QR in a shop
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NEPAL RASTRA BANK'S ROLE IN THE DIGITAL ECONOMY

- NRB regulates all **digital payment providers** (PSPs & PSO) to maintain trust and security.
 - Introduced **National Payment System Strategy** → standardizes and strengthens digital payments.
 - Ensures **safe, interoperable, and inclusive** digital financial systems.
 - Drives Nepal's move from **cash-based to cashless** economy
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WHY NRB POLICIES ARE CRITICAL FOR THE DIGITAL ECONOMY

- Build trust in digital payments
 - Prevent money laundering & fraud
 - Enable seamless online transactions
 - Support e-commerce growth
 - Encourage digital service adoption
 - Promote financial innovation
 - Expand digital inclusion to rural areas
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IT & STARTUP POLICIES

- IT Policy 2072 guides digital development
 - Startup Act (proposed) for tax benefits
 - Digital Nepal Framework → 8 sectors
 - Example: Health app startups supported
 - Encourages innovation in finance & agriculture
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IT POLICY OF NEPAL 2072 — KEY POINTS

LINK TO MORE: [IT POLICY](#)

- **Vision:** Transform Nepal into a knowledge-based society and economy.
 - **Mission:** Use ICT to drive sustainable development and reduce poverty.
 - **Digital Governance:** Make government services transparent, efficient, and inclusive.
 - **ICT Industry & Innovation:** Promote domestic software, startups, and ICT-enabled SMEs.
 - **Infrastructure & Inclusion:** Build secure, reliable ICT infrastructure and bridge the urban-rural digital divide.
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STARTUP ACT (PROPOSED) — KEY POINTS

- **Objective:** Promote entrepreneurship and ease of doing business for startups.
 - **Tax Benefits:** 5–7-year income tax break for eligible startups.
 - **Support:** Access to subsidized loans, grants, and incubation programs.
 - **Eligibility:** Startups with innovative products/services and annual turnover below a specified limit (~NPR 100M).
 - **Focus Sectors:** ICT, AI, fintech, health-tech, education, agriculture, tourism, creative industries.
 - **Goal:** Encourage innovation, employment, and export-oriented startups in Nepal.
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DIGITAL NEPAL FRAMEWORK

- **Education:** Digital learning platforms, e-schools, remote education.
 - **Health:** e-Health services, telemedicine, digital health records.
 - **Agriculture:** Smart farming, agri-tech solutions, digital supply chains.
 - **Finance:** Mobile payments, fintech innovation, financial inclusion.
 - **Tourism:** Digital promotion, e-ticketing, virtual tours.
 - **Governance:** e-Government, online public services, transparency.
 - **Commerce & Industry:** E-commerce, digital SMEs, supply chain tech.
 - **Infrastructure & Connectivity:** Broadband, 5G, national ICT backbone.
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E-COMMERCE REGULATIONS

- Consumer protection & refund rules
 - VAT on online services
 - Return/refund policy (Daraz, SastoDeal)
 - Example: Daraz “Cashback & Return” policy
 - Ensures user trust and safety
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DATA PRIVACY & CYBERSECURITY

- Privacy Act 2075 (basic protection)
 - No dedicated Data Protection Law yet
 - Cybercrime laws under Electronic Transactions Act
 - Example: App collecting location without consent
 - Gap → need stronger privacy regulations
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ETHICAL CONSIDERATIONS

AI ETHICS

- Algorithm bias (e.g., facial recognition errors)
 - Transparency: Users must know AI decisions
 - Accountability for AI-driven outcomes
 - Example: AI chatbots giving wrong suggestions
 - Ethical AI → fairness & inclusion
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PRIVACY ISSUES

- Apps track location, contacts, browsing
 - Informed consent often missing
 - Data misuse risks
 - Example: Facebook ad targeting
 - Users need awareness & control
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FAIR DIGITAL LABOR

- Gig workers (Pathao, Foodmandu riders)
 - Fair pay, insurance, safety measures needed
 - Example: Riders' accident insurance program
 - Ethics → protect human workers in digital economy
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SUSTAINABILITY IN 4IR

E-WASTE CHALLENGES

- Phones, laptops, routers, batteries
 - Nepal lacks recycling infrastructure
 - Health & environmental impact
 - Example: Old laptops in Kathmandu dumps
 - Need e-waste collection programs
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GLOBAL VS NEPAL GAP ANALYSIS

- Global: High-speed internet, 5G
 - Nepal: Slow internet, unreliable power
 - Example: Video conferencing in rural Nepal
 - Challenge: Expand broadband access
 - Opportunity: Mobile internet expansion
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AI, R&D, WORKFORCE

- Global: Large skilled workforce, R&D funded
 - Nepal: Limited AI/cloud/data skills
 - Gap → universities & startups need support
 - Example: AI farm advisory apps → underdeveloped
 - Skill development critical
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GROWTH OPPORTUNITIES

- Digital payments growth
 - AI-enabled sectors & remote monitoring
 - Digital education platforms
 - Remote work & outsourcing
 - Example: eSiksha online classes
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PRACTICAL MICRO- EXAMPLES

DAILY APP ECOSYSTEM MAPPING

- Example: Foodmandu
 - Digital: App, payments, GPS
 - Knowledge: Reviews & recommendation data
 - 4IR: Route optimization & demand prediction
 - Value: Efficient delivery & user satisfaction
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VALUE CONTRIBUTION ACTIVITY

- 1 review → boosts credibility
 - Clicks → feed algorithms
 - Time spent → platform engagement
 - Teaches data value in real life
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FUTURE SKILLS

MUST-HAVE SKILLS

- AI & ML basics
 - Data analytics & visualization
 - Cloud computing (AWS, GCP, Azure)
 - Cybersecurity fundamentals
 - Digital marketing & UI/UX
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ROLE OF UNIVERSITIES

- Update curriculum for 4IR
 - Labs & hands-on projects
 - Industry internships
 - Encourage innovation & startups
 - Example: Student AI projects for agriculture
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ONLINE LEARNING PLATFORMS

- Global: Coursera, Udemy, LinkedIn Learning
 - Nepali: eSiksha, Kullabs
 - Courses: AI, cloud, digital marketing
 - Example: Students complete ML project online
 - Enables lifelong learning
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INFLUENCE OF DIGITAL ECONOMY

SUSTAINABILITY IMPACT

- Paperless offices & digital documents
 - Smart energy systems → reduce consumption
 - Remote work → fewer transport emissions
 - Example: Digital government services in Nepal
 - Supports environmental goals
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ENERGY
STORAGE
SYSTEM

Alphaess
JT 61
9950111
2253
Madin Bhandal University
of Science and Technology



PRIVACY & REGULATION

- Increased data collection → responsibility
 - Digital taxation → VAT on services
 - Stronger e-commerce & cybersecurity rules
 - Example: e-commerce compliance with NRB
 - Protects users & builds trust
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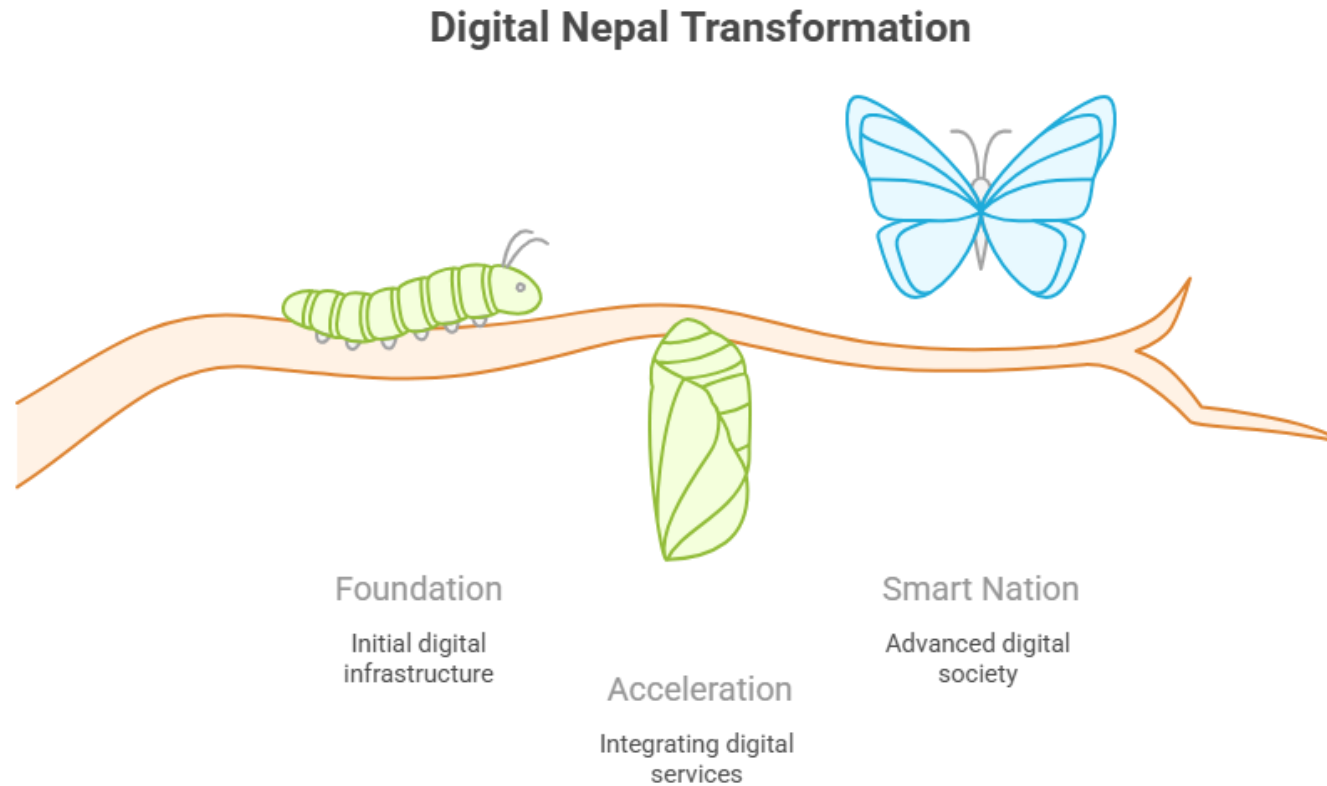
BUSINESS STRATEGY CHANGES

- Data-driven decisions → efficiency
 - Digital-first operations → cost reduction
 - Automation → reduces human errors
 - Example: Online retail inventory management
 - Competitive advantage in market
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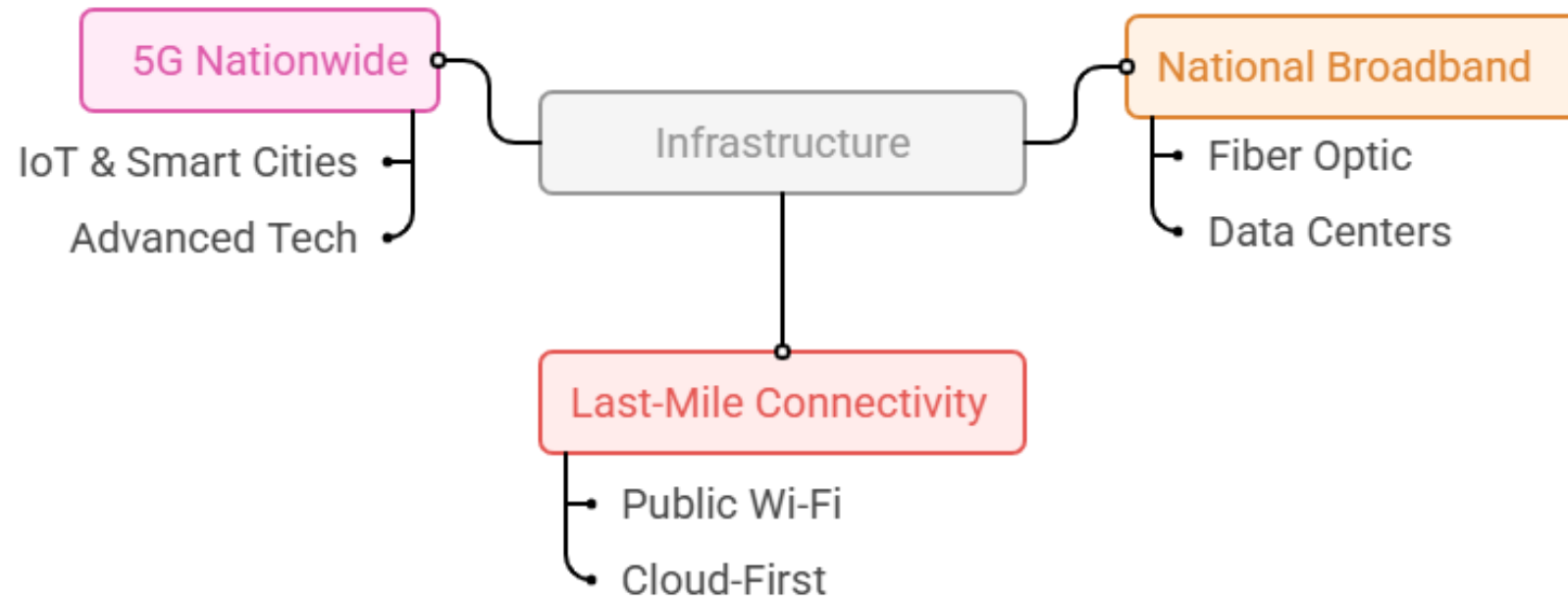
CONCLUSION & FUTURE OUTLOOK

- Recap: Policies, ethics, sustainability, skills
 - Opportunities: FinTech, AI, digital education
 - Call to action: Students prepare for digital jobs
 - Visual: Roadmap of Nepal's digital transformation
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ROADMAP OF DIGITAL NEPAL TRANSFORMATION



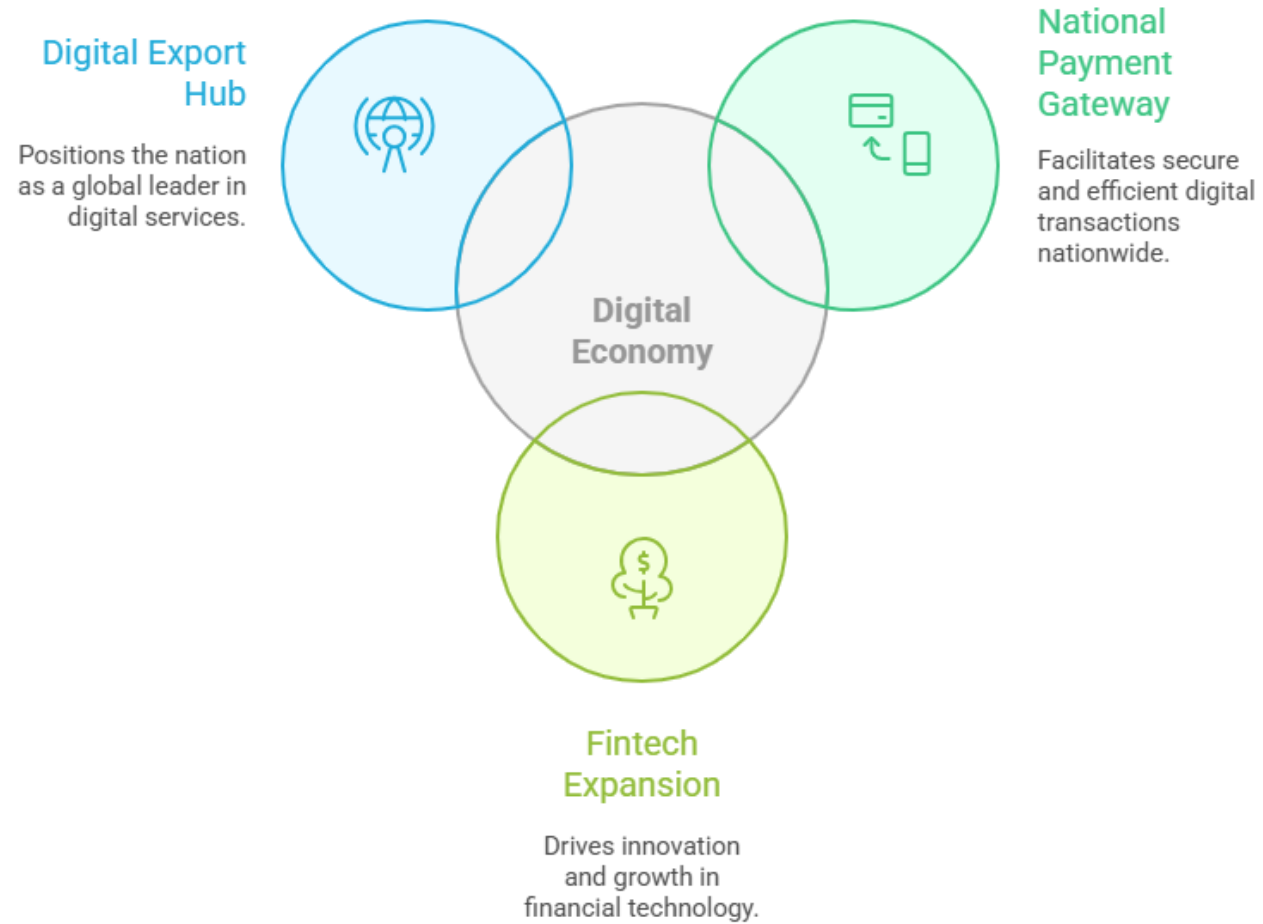
Digital Infrastructure Development in Nepal



Digital Government Transformation



Digital Economy Framework



Digital Skills Comparison

	Basic	Advanced	Lifelong
 Focus	Digital Literacy	Tech Skills	Learning
 Education	School Curriculum	Coding	AI-Powered Education
 Application	Freelancing	Incubators	Talent Hub

Digital Governance Framework

